



AMITY UNIVERSITY

AMITY INSTITUTE OF INFORMATION TECHNOLOGY

WEEKLY PROGRESS REPORT
FOR THE WEEK COMMENCING (18/05/2026-24/05/2026)

WPR No: 03

Program: MCA

NAME: DHRUV DHAYAL

ENROLLMENT NO.- A010145025215

SEMESTER: 3rd

FACULTY GUIDE NAME: Dr. Garima Bohra

PROJECT TITLE: RADMAP – GPS Enabled Radiation Mapping Prototype using Raspberry Pi

TARGETS FOR THE WEEK:

1. Research the integration of LiDAR sensors, ESP8266-CAM modules, servo motors, and Raspberry Pi for radiation monitoring applications.
2. Design the architecture of a Digital Twin-based Radiation Mapping System.
3. Study the Safecast radiation mapping platform and its data visualization framework.
4. Develop a dynamic dashboard for real-time radiation monitoring and location-based mapping.
5. Implement GPS-based radiation data collection and visualization on interactive maps.

ACHIEVEMENTS OF THE WEEK:

1. Successfully researched and analyzed the integration of LiDAR sensors, ESP8266-CAM modules, servo motors, and Raspberry Pi for environmental monitoring.
2. Developed the initial framework of a Digital Twin-based Radiation Mapping System.

3. Integrated GPS-based location tracking with radiation data collection for real-time monitoring.
4. Developed a dynamic dashboard for radiation visualization and analysis.
5. Implemented Safecast-compatible radiation mapping for displaying radiation levels on interactive maps based on geographic locations.

FUTURE WORK PLANS:

1. Complete the hardware integration and testing of LiDAR, ESP8266-CAM, servo motors, and Raspberry Pi.
2. Enhance the Digital Twin model with real-time sensor synchronization and monitoring features.
3. Improve the dashboard by adding heatmaps, analytics, and advanced visualization capabilities.
4. Expand radiation data collection across multiple locations for better mapping accuracy.
5. Prepare project documentation, GitHub repository, technical report, and research paper for publication.